

Jonathan Zhang

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EDUCATION

The University of Texas at Austin

Austin, TX

Ph.D., Computational Science, Engineering, Mathematics, *Advisors: Leszek Demkowicz, George Biros*

Present

Research Interests: Finite elements, numerical methods for PDEs, high-performance computing, nonlinear problems

M.S., Computational Science, Engineering, Mathematics

Aug 2021 – May 2023

B.S., Aerospace Engineering

Aug 2017 – Dec 2020

SKILLS

Programming Languages: Proficient in Fortran, Python, MATLAB, C++

Software Packages: NumPy, SciPy, Pandas, Mathematica, SolidWorks, ABAQUS, ParaView, Git, L^AT_EX, MS Office

EXPERIENCE

JPMorganChase

New York, NY

Quantitative Research Summer Associate – Securitized Products Group

Present

Quantitative Analytics Summer Associate – Model Risk, Governance, and Review

Jun 2025 – Aug 2025

- Analyzed model assumptions and nonlinear data treatment in a CRE prepayment logistic regression model. [Python]
- Investigated joint optimization methods for nonlinear feature transformations (e.g., PPI) by using augmented loss functions.
- Developed a neural network-based transformation of loan features (TTM, PPI) that outperformed baseline linear transformation by 100+ AIC points, revealing nonlinear prepayment acceleration dynamics while preserving explainability. [JAX]
- Communicated analysis results to business leaders and management through a series of presentations and detailed model documents.

Quantitative Analytics Summer Associate – Model Risk, Governance, and Review

Jun 2024 – Aug 2024

- Implemented a k -means model in **Athena** to cluster ZIP codes by land share and RVB data to evaluate CRE MFL market strength.
- Identified several technical and methodological flaws in the initial classifier, impacting $\sim 10\%$ of the entire CRE portfolio.
- Back-tested classifier using downstream default rate (PD) and loss (LGD) models using custom performance metrics. [Python]

Oden Institute for Computational Engineering and Sciences

Austin, TX

Graduate Research Assistant – Electromagnetics and Acoustics Group

Aug 2022 –

- Exploring new applications using neural networks based on variationally stable problem formulations in biomechanics problems.
- Studying various variational formulations for non-linear elasticity including those using the Discontinuous Petrov-Galerkin (DPG) method to deliver optimal convergence rates with adaptive mesh refinement. [Fortran]

Sandia National Laboratories

Remote

Member of Technical Staff – Component Science and Mechanics

Jan 2021 – Aug 2021

- Analyzed the influence of structural dynamics on electrical chatter in a pin-receptacle using high- and low-fidelity models in SIERRA.
- Developed a Craig-Bampton reduced-order model for the pin-receptacle, reducing compute time by 30X on Sandia's HPC's.

Oden Institute for Computational Engineering and Sciences

Austin, TX

Moncrief Research Intern

Mar 2019 – Aug 2019

- Simulated thin film interfacial properties in a double cantilever beam under tension with different bonding strengths using ABAQUS.
- Compared finite element results with self-gathered experimental data to determine significance of imperfect bonding.
- Formulated a polynomial model for imperfect bonding conditions based on ABAQUS results to within 0.2% error.

SELECTED PUBLICATIONS

- J.Z.** and Leszek Demkowicz, “Construction of a Robust Formulation for Nearly Incompressible Nonlinear Elasticity,” *Computers and Mathematics with Applications*, Submitted 2026.
- J.Z.** and Leszek Demkowicz, “Nonlinear Elasticity with the DPG Method. The Steepest Descent Method,” *Computer Methods in Applied Mechanics and Engineering*, 454:118875, 2026
- J.Z.** and Leszek Demkowicz, “Nonlinear Elasticity with the DPG Method I. Various Variational Formulations,” *Computers and Mathematics with Applications*, 200:276-289, 2025